

# Jack Petok

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US Citizen

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## Employment

2024-2026 *Visiting Assistant Professor*, Colby College, Waterville ME

## Previous positions

2020-2024 *Lecturer*, Dartmouth College, Hanover NH

Sept-Dec 2023 *Guest Researcher*, Hausdorff Institute for Mathematics, Bonn, Germany

## Research interests

Algebraic geometry; arithmetic geometry; moduli spaces; derived categories;  $K3$  surfaces and hyperkähler varieties; rational points.

## Education

2011-2015 BS in Mathematics with Honors, Stanford University

2015-2020 PhD in Mathematics, Rice University (Advised by Anthony Várilly-Alvarado)

## Grants, honours & awards

2017 NSF Graduate Research Fellowship Program Honorable Mention

## Publications & preprints

Papers available on webpage: <https://jackpetok.github.io/index.html#research>

Kodaira dimension of moduli of special  $K3^{[2]}$ -fourfolds of degree 2 (Journal des Mathématiques Pures et Appliquées, Neuvième Série 173 (2023), 149–171.)

A census of cubic fourfolds over  $\mathbb{F}_2$ , with Asher Auel, Avinash Kulkarni, and Jonah Weinbaum (Mathematics of Computation, 94 (2025), 2089-2112).

On decompositions for Fano schemes of intersections of two quadrics, with Pieter Belmans, Jishnu Bose, Sarah Frei, Benjamin Gould, James Hotchkiss, Alicia Lamarche, Cristian Rodriguez Avila, and Saket Shah (accepted 2025 at Advances in Mathematics).

Zeta functions of  $K_3$  categories over finite fields, with Asher Auel (submitted in 2025).

Rational points on modular curves of composite level, with Catalina Camacho Navarro, Wanlin Li, Jackson Morrow, and David Zureick-Brown (Preprint).

Geometric realizations of Brauer classes from hyperkähler geometry (in preparation).

Sections of quadric fourfold bundles over surfaces, with Asher Auel (in preparation).

## Teaching

Some course webpages: <https://jackpetok.github.io/index.html#teaching>

### TEACHING AT COLBY

Math III (Mathematics as a Liberal Art), Spring 2026

Math 130 (Calculus I Revisited), Spring 2026

Math 160 (Series and Multivariable Calculus), Fall 2025

Math 130 (Calculus I Revisited), Fall 2025

Math 130 (Calculus I Revisited), Spring 2025

Math 160 (Series and Multivariable Calculus), Fall 2024

Math 130 (Calculus I Revisited), Fall 2024

### TEACHING AT DARTMOUTH

Math 8 (Calculus of functions of one and several variables), Spring 2023

Math 3 (Calculus), Winter 2023

Math 9 (Multivariable Calculus with Linear Algebra), Fall 2022

Math II (Accelerated Multivariable Calculus), Fall 2022

Math 8 (Calculus of functions of one and several variables), Spring 2022

Math 8I/Math III (Galois Theory) Winter 2022

Math 25 (Number Theory) Fall 2021

Math 31 (Topics in Algebra) Fall 2021

Math 22 (Linear Algebra with Applications) Spring 2021

Math 8 (Calculus of functions of one and several variables), Winter 2020

Math 22 (Linear Algebra with Applications) Fall 2020

Math 8 (Calculus of functions of one and several variables), Fall 2020

## TEACHING AT RICE

Math 365 (Elementary Number Theory), Summer 2020

Qualifying exam course for graduate students, Summer 2020

Math 355 (Linear algebra), Summer 2018

Math 211 Ordinary Differential Equations and Linear Algebra, Fall 2017

## Advising

Advisor to Dartmouth senior thesis student Jonah Weinbaum (co-advised with Asher Auel and Avinash Kulkarni, 2022-23)

Thesis reader and co-advisor to Colby student Vladimir Khabaev (co-advised with Fernando Gouvêa, 2024-25)

Independent study mentor to Colby students (2025-26)

## Talks

Geometric realizations of Brauer classes from hyperkähler geometry (January 2026, Rice University Algebra and Number Theory Seminar, Houston TX)

Geometric realizations of Brauer classes from hyperkähler geometry (January 2026, Special Session on Hyperkähler Varieties at Joint Mathematics Meetings, Washington DC)

Geometric realizations of Brauer classes from hyperkähler geometry (October 2025, AGPUI Virtual Seminar)

Geometric realizations of Brauer classes from hyperkähler geometry (Maine Geometry and Topology Mini-conference, 2025, Waterville, ME)

Quadratic forms and lattices (October 2024, Colby College Colloquium, 2024, Waterville, ME)

The zeta function of a  $K3$  category (October 2023, Junior Trimester Program, Bonn, Germany)

A survey of cubic fourfolds over  $\mathbb{F}_2$  (April 2023, South Carolina Algebra, Geometry, and Number Theory Seminar, Columbia, SC)

A survey of cubic fourfolds over  $\mathbb{F}_2$  (April 2023, Dartmouth Algebra and Number Theory Seminar)

A census of cubic fourfolds over  $\mathbb{F}_2$  (January 2023, JMM Special Session on Arithmetic geometry informed by computation, Boston, MA)

Composite level Galois representations and low genus modular curves (AMS Special Session on Algebraic and analytic methods in the theory of elliptic curves, Amherst MA)

Lightning talk: spaces of sections of quadric bundles (July 2022, Banff Workshop on Derived categories and reconstruction in algebraic geometry)

Spaces of sections of quadric bundles (June 2022, Dartmouth Algebra and Number Theory Seminar)

Spaces of sections of quadric bundles (December 2021, Michigan State Algebraic Geometry Seminar)

Composite level Galois representations and low genus modular curves (October 2021, Dartmouth Algebra and Number Theory Seminar).

Kodaira dimensions of some moduli spaces of hyperkähler fourfolds (October 2021, Algebraic Geometry Northeastern Series (AGNES))

Kodaira dimensions of some moduli spaces of hyperkähler fourfolds (March 2021, Algebraic Geometry Seminar at University of British Columbia)

Kodaira dimensions of some moduli spaces of hyperkähler fourfolds (Fall 2020, Dartmouth Algebra and Number Theory Seminar)

Kodaira dimensions of some moduli spaces of hyperkähler fourfolds (October 2020, AMS Fall Southeastern Sectional Meeting - AMS Special Session on Geometry and Arithmetic of Hyperkähler Manifolds)

Composite level Galois representations and low genus modular curves (June 2020, Chicago Number Theory Day).

Kodaira dimension of moduli spaces of low degree special  $K3^{[2]}$ -fourfolds. and Computation (January 2020, Joint Mathematics Meetings - AMS Special Session on Rational Points on Algebraic Varieties: Theory)

Kodaira dimension of moduli spaces of low degree special  $K3^{[2]}$ -fourfolds (October 2019, Algebra Seminar at Brown University)

Kodaira dimension of moduli of special  $K3^{[2]}$ -fourfolds of degree 2 (May 2019, , Reinventing Rational Points at Institut Henri Poincaré)

Sphere packings in dimensions 8 and 24 (February 2019, , Rice Undergraduate Mathematics Colloquium)

## **Organization and service to the profession**

Co-organizer of Colby College Putnam Team (2025-26)

Co-organizer of seminar on Selected Topics Relating To Cubic and Hyperkählers (co-organized with Xuqiang Qin, 2023, Bonn)

Co-organizer of AMS Special Session on Brauer Groups in Algebraic Geometry and Arithmetic (co-organized with Sarah Frei, 2023, Cincinnati)

Co-organizer of Dartmouth Brauer Group Reading Seminar (co-organized with Richard Haburcak, 2022, Dartmouth)

Co-organizer of Student Number Theory Seminar, (2015—2019, Rice)

Co-organizer of Video Seminar, (2018—2019, Rice)

Departmental Graduate Representative, Gulf Coast Undergraduate Research Symposium, (2019, Rice)

Volunteer and Speaker, Rice MathExplorations and Rice Mathematics Circle, (2015-2019, Rice)

## **Referee and review work**

Revista Matemática Iberoamericana, Documenta Mathematica, zbMath, MathSciNet

Last updated: September 7, 2025